Linear Regression

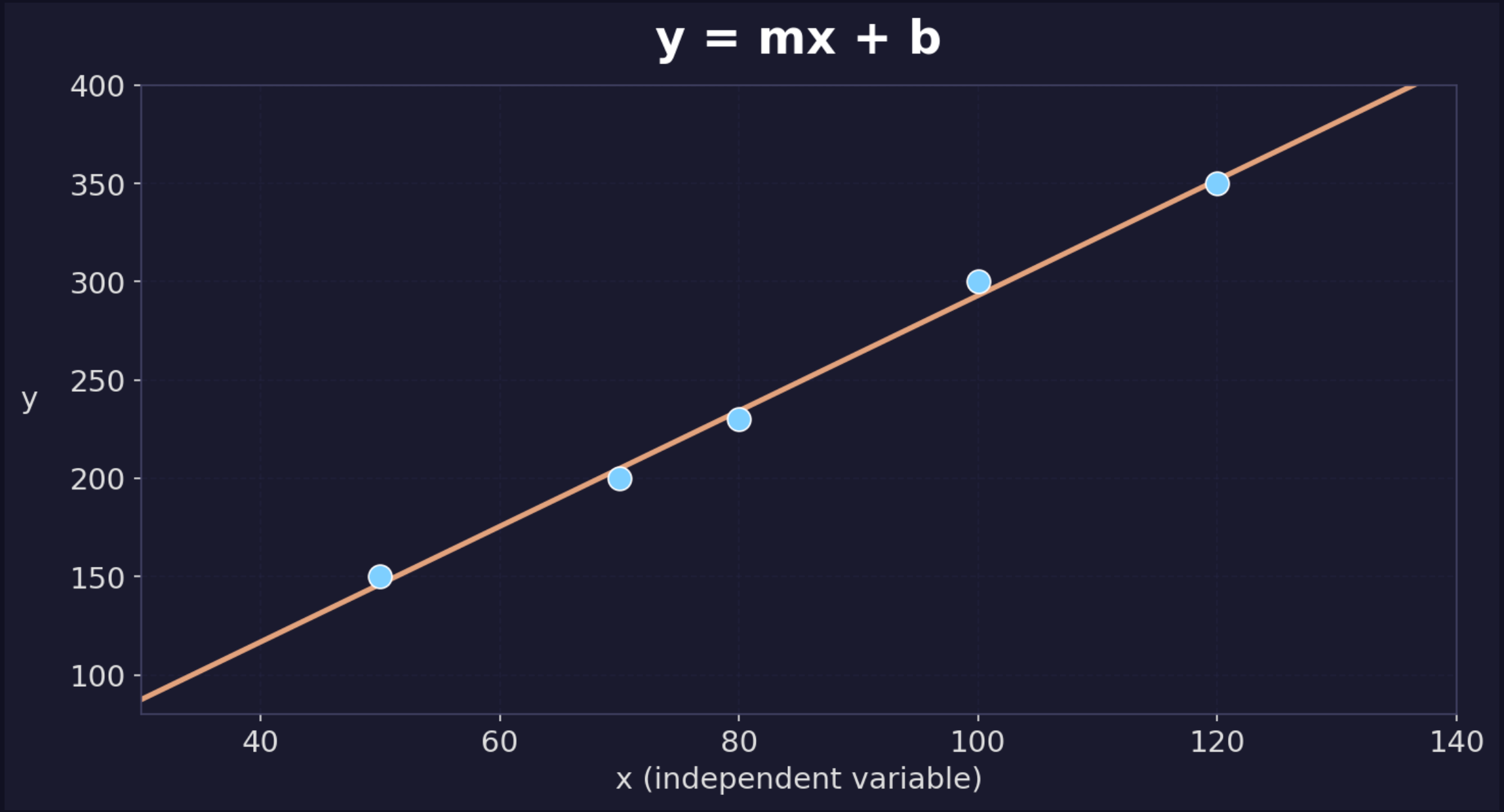
A supervised learning algorithm that models the relationship between a dependent variable and one or more independent variables by fitting a straight line.

$$y = mx + b$$

m = slope · **x** = independent variable · **b** = y-intercept · **y** = predicted value

$$m = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$$b = \bar{y} - m\bar{x}$$

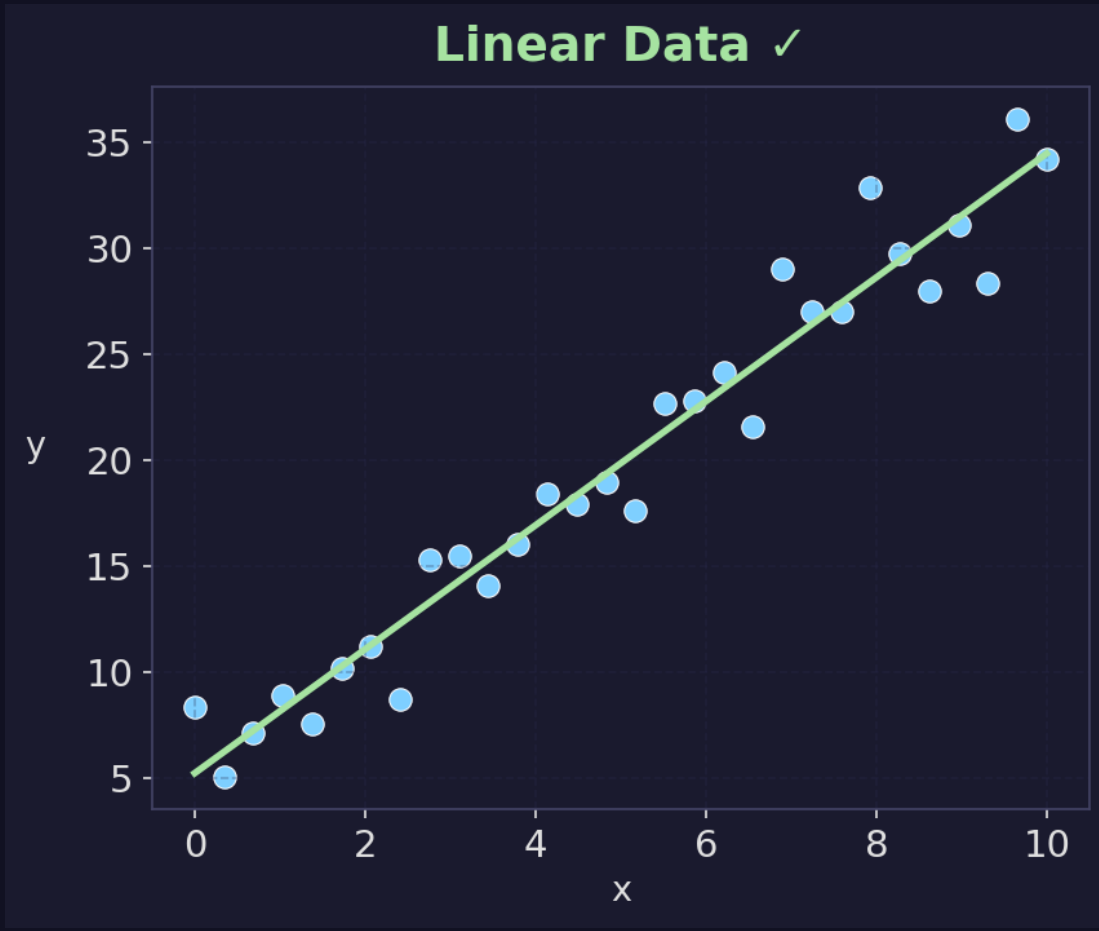

y = mx + b
400
350
300
250
200
150
100
y
40
60
80
100
120
140
x (independent variable)

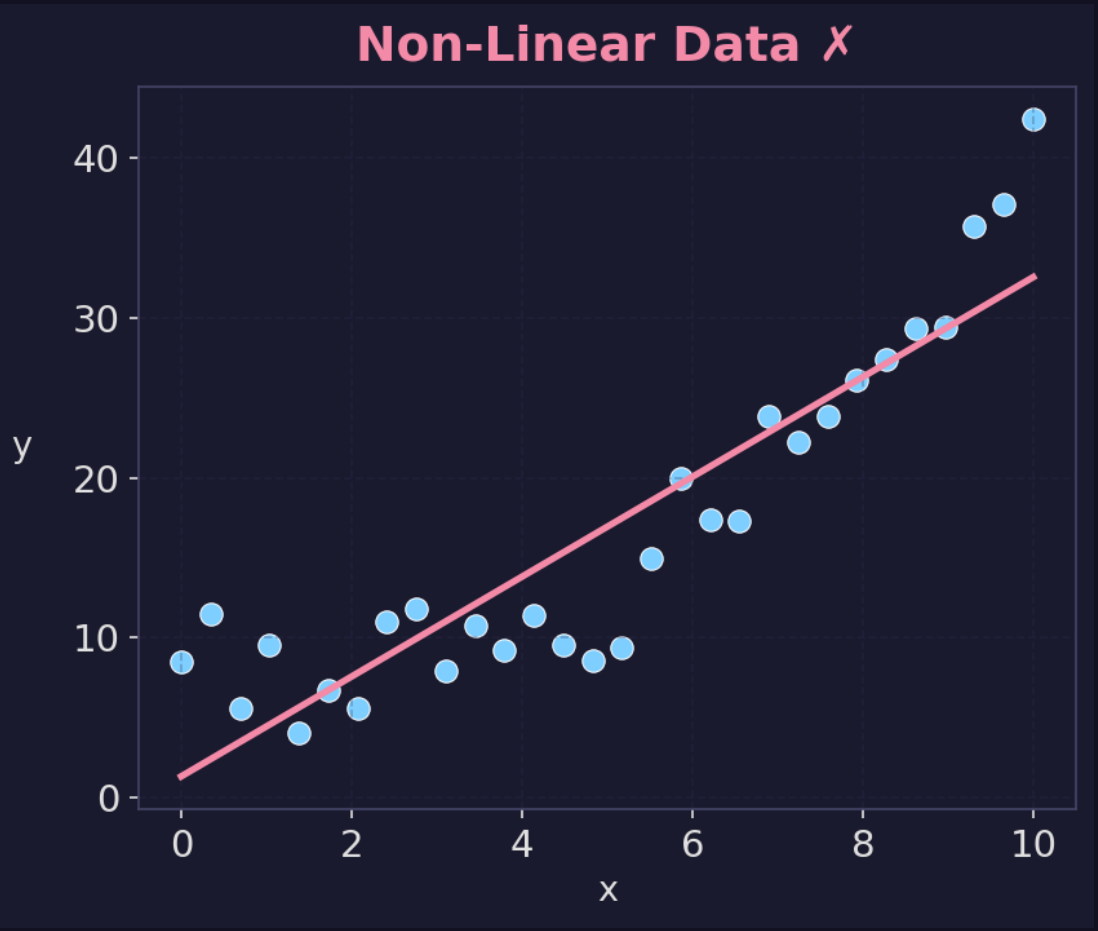
Pros

- **Simple & Interpretable** — coefficients directly explain feature impact
- **Computationally Efficient** — scales well to large datasets
- **Strong Baseline** — quick to implement, easy to benchmark against
- **Well-understood Theory** — backed by solid statistical foundations
- **Low Variance** — stable predictions across similar datasets

Cons

- **Assumes Linearity** — cannot capture curved or complex relationships
- **Sensitive to Outliers** — a few extreme points can skew the entire line
- **Multicollinearity Issues** — correlated features degrade coefficient reliability
- **Underfitting Risk** — too simple for most real-world problems on its own
- **Requires Feature Engineering** — raw features often need transformation to work well


Linear Data ✓
35
30
25
20
15
10
5
y
0
2
4
6
8
10
x


Non-Linear Data ✗
40
30
20
10
0
y
0
2
4
6
8
10
x

Use Case: Real Estate Price Prediction

Problem: Predict house prices based on square footage.

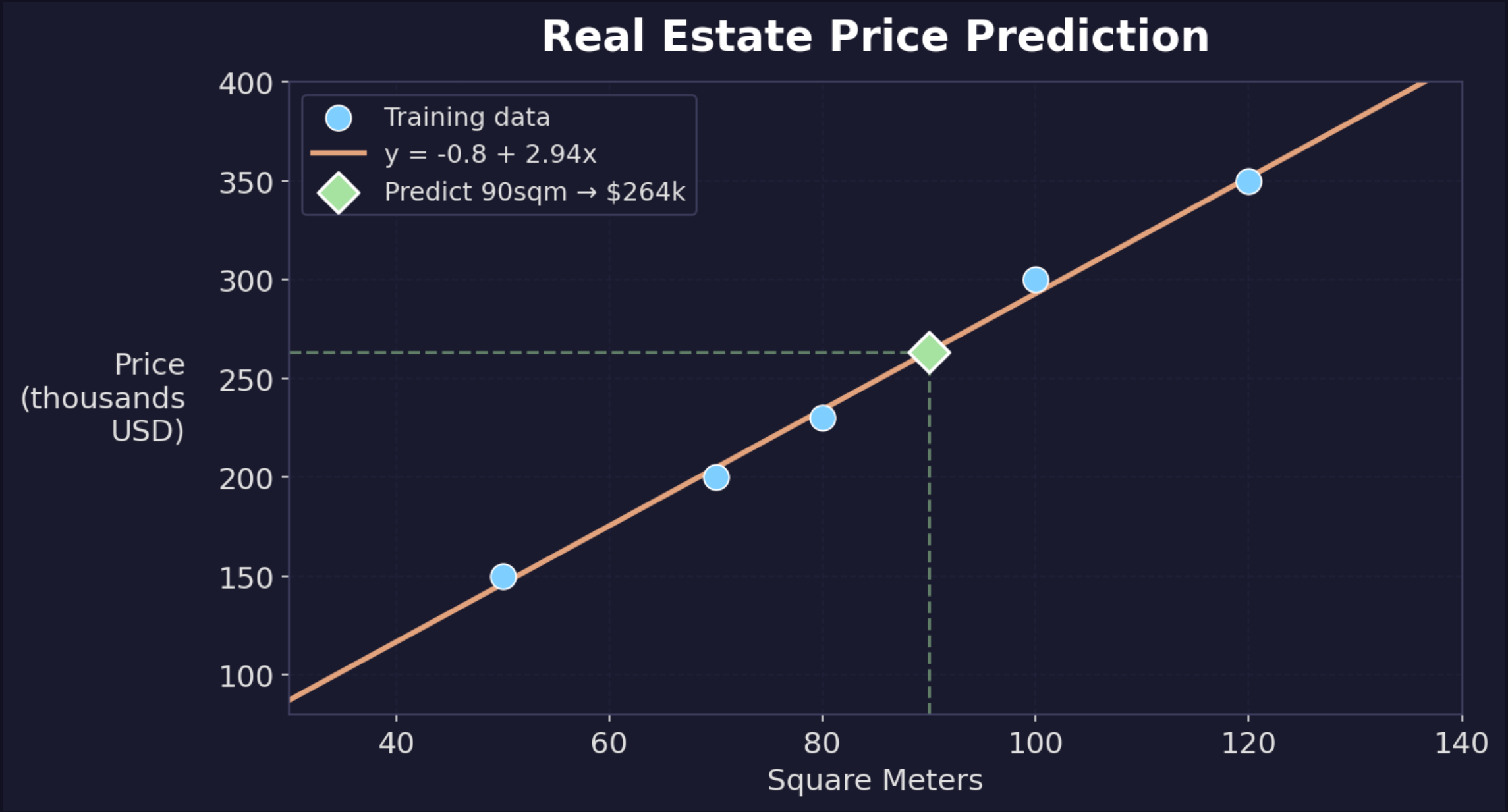
Square Meters	Price (USD)
50	150,000
70	200,000
80	230,000
100	300,000
120	350,000

The model learns a line:

$$price = b + m \cdot sqm$$

Then predicts the price for any new size.

Other applications: stock forecasting, crop yield estimation, sales analysis.


Real Estate Price Prediction
Training data
y = -0.8 + 2.94x
Predict 90sqm → $264k
Price (thousands USD)
400
350
300
250
200
150
100
40
60
80
100
120
140
Square Meters

What Is the Slope?

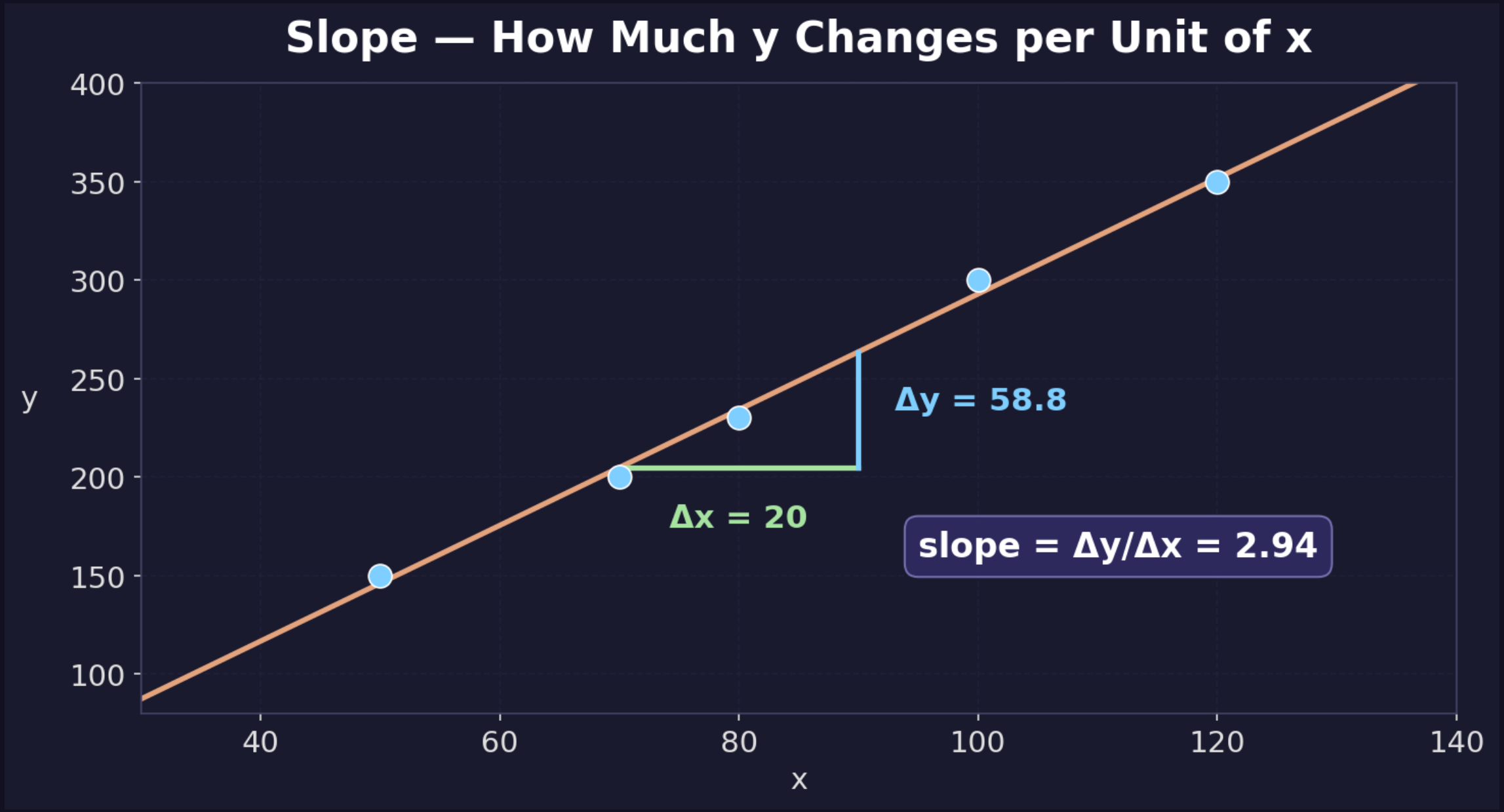
The slope **m** is *how much y changes when x increases by 1*.

$$m = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

In this dataset: slope = **2.86** — each additional square meter adds **$2,860** to the price.

The intercept **b** is where the line crosses y when x = 0:

$$b = \bar{y} - m\bar{x}$$


Slope — How Much y Changes per Unit of x
400
350
300
250
200
150
100
y
40
60
80
100
120
140
x
Δy = 58.8
Δx = 20
slope = Δy/Δx = 2.94

Code Live Demo

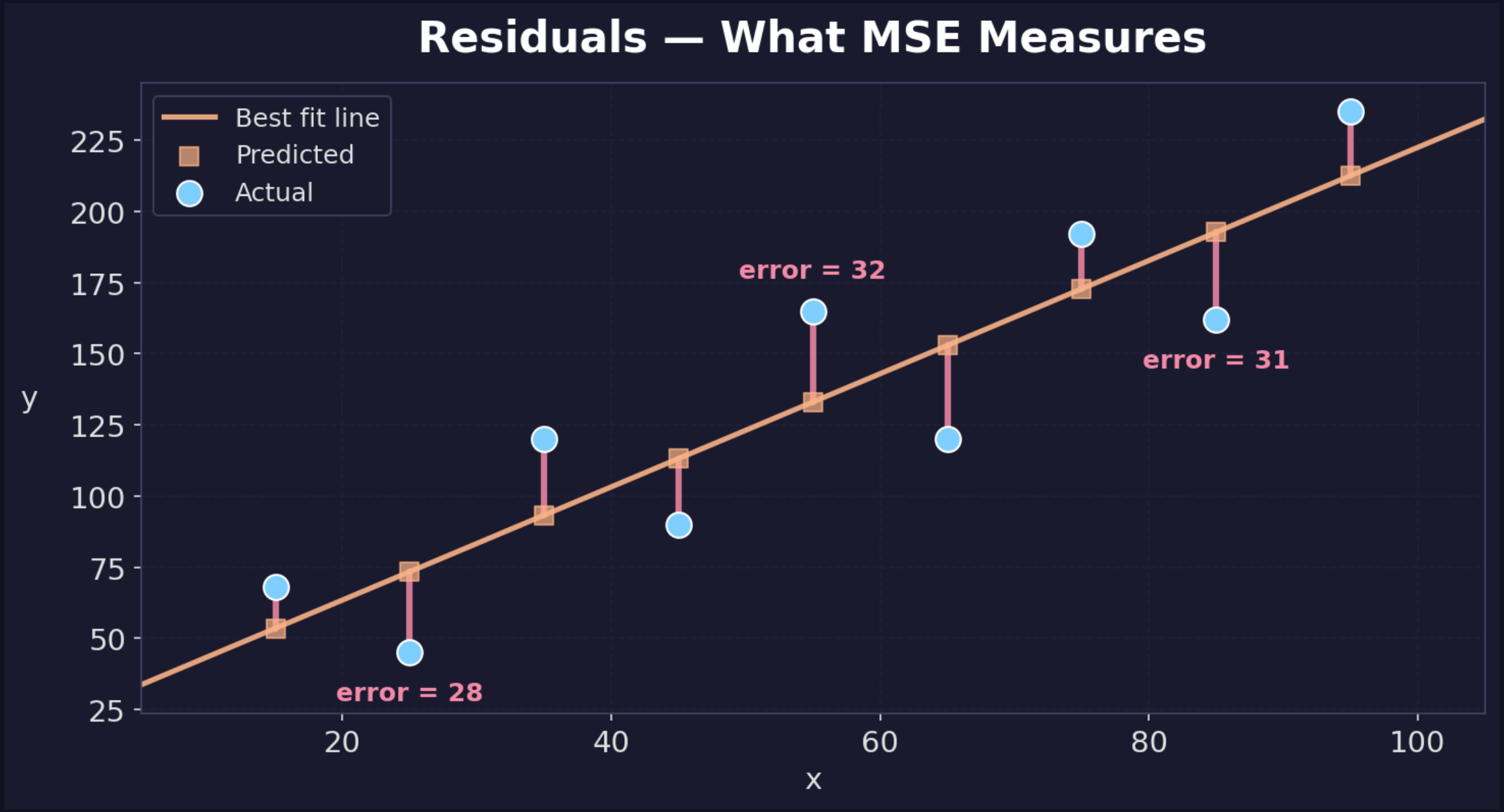
MSE — Mean Squared Error

Measures how far off predictions are, on average.

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

1. For each point: **error** = actual - predicted
2. **Square** each error (penalizes big misses harder)
3. **Average** them all

Lower MSE = better fit. MSE of **0** = perfect predictions.


Residuals — What MSE Measures
Best fit line
Predicted
Actual
error = 32
error = 31
error = 28
y
x
225
200
175
150
125
100
75
50
25
20
40
60
80
100

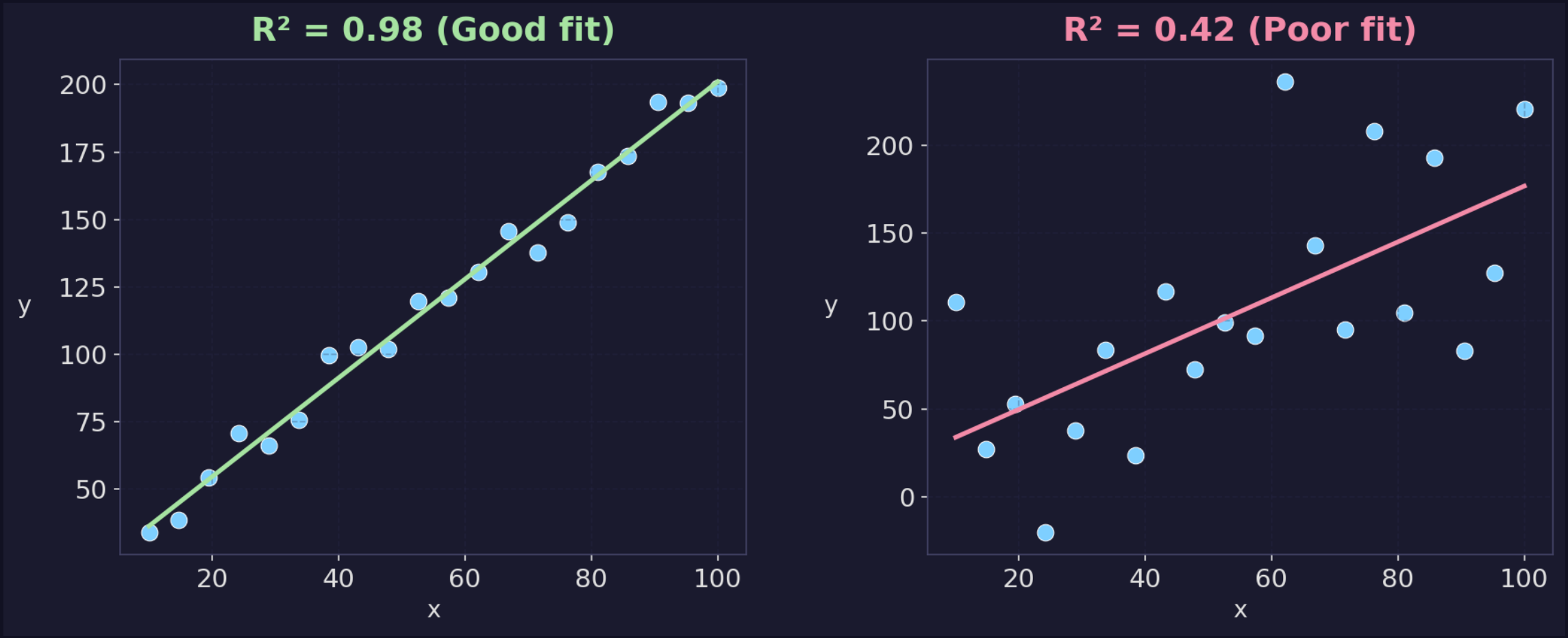
R^2 — How Good Is the Fit?

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

R^2 = 1.0 → perfect fit (predictions match exactly)

R^2 = 0.0 → no better than predicting the mean

Aspect	Detail
Type	Supervised Learning — Regression
Formula	y = mx + b
Best for	Linear relationships, quick baselines
Avoid when	Complex, non-linear patterns


R² = 0.98 (Good fit)
200
175
150
125
100
75
50
y
20
40
60
80
100
x
R² = 0.42 (Poor fit)
200
150
100
50
0
y
20
40
60
80
100
x

Thank You